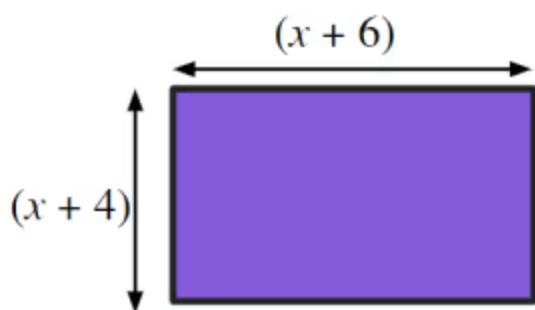


Problem solving with expressions and formulae

1 What do you call a shape which is created using two or more basic shapes? Tick **1** correct answer

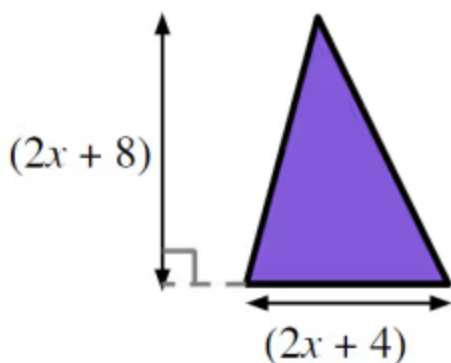
- ☒ a compound shape
- ☐ an irregular shape
- ☐ a parallelogram
- ☐ a polygon
- ☐ a trapezium

2 Which of these is an expression for the area of this rectangle? Tick **1** correct answer



- ☐ $2x + 10$
- ☐ $4x + 20$
- ☐ $x^2 + 24$
- ☐ $x^2 + 10x + 10$
- ☒ $x^2 + 10x + 24$

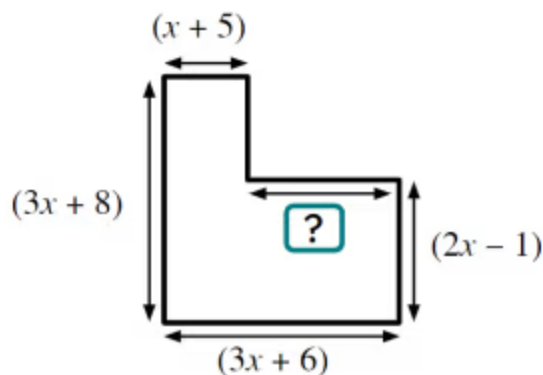
3 Which of these is an expression for the area of this triangle? Tick **1** correct answer



- ☐ $x^2 + 6x + 8$
- ☐ $2x^2 + 6x + 16$
- ☐ $2x^2 + 10x + 16$
- ☒ $2x^2 + 12x + 16$
- ☐ $4x^2 + 24x + 32$

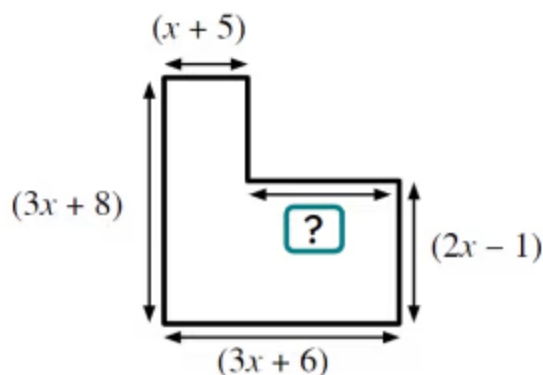


- 4 Find an expression for the horizontal length marked with a '?' on this compound shape. Tick **1** correct answer



- ☐ $x + 7$
☐ $x + 9$
☒ $2x + 1$
☐ $2x + 11$
☐ $4x + 11$

- 5 Which of these is an expression for the area of this compound shape? Tick **1** correct answer



- ☐ $5x^2 + 19x + 41$
☐ $7x^2 + 13x + 12$
☒ $7x^2 + 23x + 39$
☐ $7x^2 + 33x + 53$
☐ $7x^2 + 61x + 57$

- 6 All the length marked on this diagram are in centimetres. The area of the shaded part of this shape is 102 cm^2 . The value of x is 5. Fill in the blank

